

Behavioral timescale synaptic plasticity: properties, elements and functions

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Jeffrey C. Magee ^{1,2} 

Understanding how brains learn and remember remains among the most important challenges in science. Recent studies in the hippocampus implicate a new form of synaptic plasticity, named behavioral timescale synaptic plasticity (BTSP), in the generation of experience-based learning and memory. BTSP is a strong, bidirectional type of plasticity that affects synaptic weights over many seconds of time. It is induced by single dendritic plateau potentials, as opposed to many action potentials, and is thus capable of producing new place cells in one trial. Plateau potential initiation is controlled, at least in part, by local feedback inhibition and an instructive input from a higher-order brain region that potentially links the plasticity to current experience. The new credit assignment procedure in BTSP provides a nonstandard mechanism for memory storage and retrieval that could mitigate the need for widespread synapse stabilization. In addition, it may allow hippocampal networks both to form memories of specific behavioral episodes and to generalize on the basis of past episodes. Finally, recent BTSP investigations could provide a basis for future explorations into how brains learn and remember, ranging from the systems and cognitive levels down to the basic biochemical building blocks of learning and memory.

One of the main goals of neuroscience is to understand how brains learn and remember just by interacting with the environment. Examples of this learning and memory range from how we find our way home from work every night to something more exotic, such as how we retain the individual episodes of our lives that are so important for our unique sense of self. The centrality of this fundamental cognitive function for human well-being is made clear by various brain disorders that degrade learning and memory and, albeit to a lesser extent, even by normal aging.

Over the past few decades, many labs have worked to understand learning and memory, and recent discoveries point to new cellular and circuit-level mechanisms that may be involved. In this Review, I discuss how breakthroughs in our understanding of dendritic integration, synaptic plasticity and neuronal microcircuit function have led to a new appreciation of the neuronal substrates of learning and memory. In addition, I highlight work on the molecular and cell biological mechanisms of relevant synaptic plasticity and discuss areas where new experiments can close gaps in our current knowledge. Future work

in the coming decades promises to produce substantial advances in our understanding of how brains learn and remember under both normal and pathological conditions.

Learning and memory as neuronal adaptations to the environment

As a simple example of learning from experience, consider what happens in the brain of a mouse when it encounters a new environment (Fig. 1a). Initially, the mouse does not know the best route through the environment, but as it explores, adaptive changes occur within its brain. These changes are adaptive in the sense that they improve the behavior of the mouse such that, after some exploration, the brain can produce the best route through the environment. If the mouse returns to this environment a few weeks later, a portion of the brain changes will have been maintained, and these allow the brain to generate a satisfactory route. What is being broadly described here is a two-stage process. The first stage is the formation of adaptive changes that will be, for

¹Howard Hughes Medical Institute, Baylor College of Medicine, Houston, TX, USA. ²Jan and Dan Duncan Neurological Research Institute, Texas Children's Hospital, Houston, TX, USA. ✉e-mail: jcmagee@bcm.edu

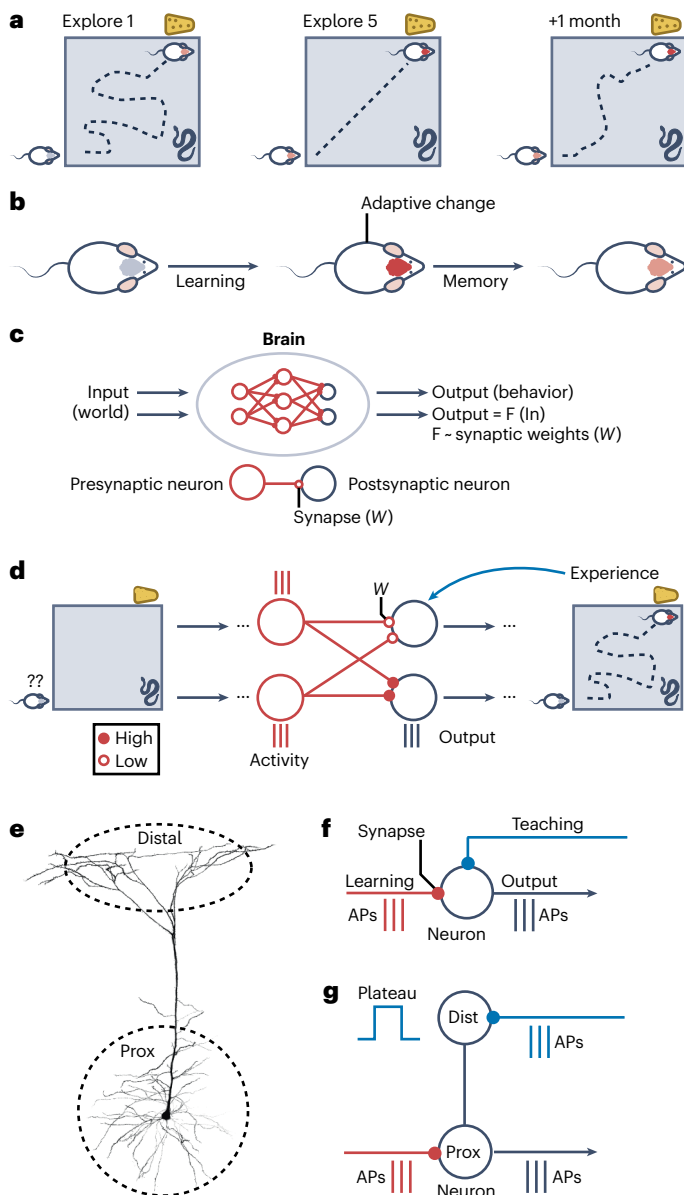


Fig. 1 | Learning and memory in brains. **a**, From left, the trajectory (dashed line) of a mouse during the initial exploration of a new environment. The development of an adaptive change in the mouse's brain (increased opacity of red oval) improves the trajectory after five explorations. The reward location is indicated by the cheese, and a negative valence region is indicated by the snake. One month later, the mouse re-encounters the environment, and remaining adaptive change produces a good route. **b**, Development and partial maintenance of adaptive change. **c**, Sketch of the brain (network of neurons connected by synapses) transforming input (left red neurons) from the environment into output (black neurons) that drives behavior. Function is partially determined by synaptic weights connecting neurons. **d**, A similar sketch illustrating the link between the behavioral experience of the mouse that directs adjustment of synaptic weights and the transformation toward improving behavior. **e**, Biophysical properties of neurons allow a potential link to experience. A pyramidal neuron is shown with its major dendritic compartments indicated (distal (dist), dashed oval; proximal (prox), dashed circle). **f**, Single-compartment neuron (black circle) receiving 'teaching' and 'learning' input with a single voltage signal (AP output). **g**, Two-compartment neuron where the distal compartment receives teaching input and has its own large-amplitude, long-duration voltage signal (plateau), and the proximal compartment receives learning input that drives AP output.

our purposes here, called learning (Fig. 1b). The second stage is the maintenance or stabilization of these initial changes that will be, for simplicity's sake, called memory.

Synaptic plasticity

It has been thought for nearly a century that adaptive changes are produced by modifications in the strength of synaptic connections between neurons, that is, synaptic plasticity^{1–7}. The idea is that synaptic plasticity adjusts synaptic weights such that the neuronal network in which they are embedded can transform a given input from the world into an activity output pattern that produces the most appropriate behavior (Fig. 1c). In this case, the adaptations mediated by synaptic plasticity are simply modified transformations that improve behavior. It should be noted, however, that although this mechanism is widely accepted, there remain skeptics of the synaptic plasticity hypothesis of learning and memory, particularly around the idea that long-term memories can be stored in highly stabilized synaptic weight changes^{8–10}.

One of the issues with learning via synaptic weight changes is that the plasticity mechanisms that have been proposed seem ill-suited to handle the task. To begin to illustrate this point, consider that the rodent hippocampal CA1 and CA3 areas contain ~500,000 principal neurons and 10 billion excitatory synapses¹¹. Although this is a lot of processing power, it raises the question of how it is possible for synaptic plasticity mechanisms to find the right synapses and appropriately adjust them to improve behavior in as short a time as possible. Even this understates the difficulties, because there are also tens of billions of synapses and other parameters between the hippocampus and the behavioral action, and these elements may also, somehow, need to be taken into account when adjusting the weights of CA1 synapses to improve behavior.

This complexity is known as the 'credit assignment problem', and as far as learning from experience through synaptic plasticity goes, there are a few issues to be considered^{1,12–15}. First, there should be some link from the animal's experience to the synaptic plasticity mechanism that guides the plasticity in producing whatever neuronal activity pattern improves behavior (Fig. 1d). Also, as unknown parts of the environment are potentially dangerous, the required exploration should be as limited as possible; thus, the plasticity needs to be capable of being induced as quickly as possible, perhaps even within a single event (one-shot learning). Finally, the plasticity should adjust synaptic weights over the appropriate timescales, which for behavior is many seconds, at the minimum.

What is known about synaptic plasticity? Since 1949, the form of synaptic plasticity thought to underlie learning and memory in brains has been Hebbian plasticity. Hebbian plasticity is based on correlations between the presynaptic and postsynaptic neuronal activity already present in the network^{2,6}. In the quintessential form of Hebbian synaptic plasticity (spike-timing-dependent plasticity), the weights of the synapses connecting two neurons are adjusted according to the time interval between the presynaptic and postsynaptic neuronal action potential (AP) activity, within a time course on the scale of tens of milliseconds^{6,16–19}. Neuronal activity is stochastic, and to protect against the learning of noise, Hebbian plasticity requires many dozens or even hundreds of presynaptic and postsynaptic pairings to produce a substantial weight change. This intriguing idea has generated a substantial amount of research over the decades, from which we have learned much about the fundamental properties of synapses and that forms of Hebbian plasticity may be particularly important for the development of fundamental network properties^{6,20–22}. However, because there is no obvious link to experience, and because the weight changes are based on correlations alone, classical Hebbian plasticity types tend to be quite slow, requiring many repeated activity pairings, operate over too limited a timescale and struggle to produce certain transformations⁶. For example, Hebbian plasticity cannot convert a silent neuron into an active one, even if experience says this would improve behavior.

Dendrites provide multiple integrative compartments

A form of plasticity that replaces the dependence on correlated activity with a type of experience-dependent teaching signal should have

enhanced capabilities for improving behavior^{23,24}. Indeed, many theoretical modifications have been proposed to bring these capabilities to classical Hebbian plasticity⁶. However, the standard 20th century view of neurons, on which Hebbian plasticity forms were based, was that neurons are single integrative compartments that possess only one large voltage signal—the AP (Fig. 1f). This is a problem for the implementation of directed forms of plasticity, because both the teaching signal and learning inputs are mixed within this single integrative compartment, and there is only one, stochastic, voltage signal for both encoding–transmitting information and for directing plasticity. Fortunately, one look at a real neuron (Fig. 1e), and a lot of additional research into this topic^{25–46}, tells us that this 20th century view is rather impoverished. Instead, neurons are highly ramified structures with dendrites that receive most, if not all, of the tens of thousands of excitatory synaptic inputs onto a given cell. Pyramidal neurons, in particular, possess two broad dendritic regions: one that surrounds the axonal output sites of neurons (proximal or perisomatic compartment), and another that is as far away from the output site as is possible (distal or apical tuft compartment; Fig. 1g). Each of these zones receives different excitatory synaptic inputs from different brain areas^{46–49}, and the electrical properties of dendrites is such that input into one zone has a limited impact on the other owing to both passive and active dendritic filtering^{25–46}. In addition, each of these zones is associated with a different large-amplitude voltage signal. The proximal inputs drive the AP output of the neuron, and APs are brief voltage spikes (~1 ms in duration) that propagate relatively weakly into dendritic arbors^{25–27,29,30,46}. The distal input, on the other hand, drives dendritic calcium plateau potentials (plateaus; Box 1) that are hundreds of milliseconds in duration, and this slowness allows them to propagate well throughout the entire neuron, where in vitro work has shown plateaus to be robust inducers of synaptic plasticity^{31,34–37,40–46,50–53}. So real neurons appear to have evolved mechanisms to separate the information encoding–transmitting elements (proximal compartment) from the plasticity-directing elements (distal compartment), and this arrangement may allow neurons to implement more powerful directed forms of synaptic plasticity^{54–56}.

Adaptive changes in hippocampus

Past research has provided solid evidence that the hippocampus is involved in both learning and memory formation^{57–61}. The hippocampus is home to the world-famous place cell (PC), and these pyramidal neurons show an abstract feature selectivity known as a place field (PF), which is where the AP activity of a given neuron increases and then decreases as an animal moves through a particular location in the world^{62,63} (Fig. 2a). As a population, PCs produce activity maps of given environments that take some minutes to develop from initial exposure to the environment^{64–66}. Hippocampal area CA1 has been repeatedly observed to form a substantial adaptive change as rodents explore a new environment (real or virtual; Fig. 2b). Here, as an animal explores, a much higher density of PCs is developed around the salient regions of the environment (reward sites or predictive cues), greatly increasing PF activity at these locations^{66–71}. Manipulations that specifically alter the formation of this PC overrepresentation, or just CA1 activity in general, have been shown to modify the animal's ability to learn^{69,71–73} (Fig. 2c,d). Thus, the experience-dependent development of this population activity, with the twofold to threefold increased density of PCs at salient regions, is truly an adaptive change that, as such, offers a unique window on the neuronal mechanisms of learning itself.

In CA1 pyramidal neurons, the proximal integrative compartment receives excitatory synaptic input from the upstream hippocampal region (area CA3)⁶, whereas the excitatory input to the distal region originates from a completely different brain region, the entorhinal cortex (EC; EC3, layer 3 neurons)^{47,74}. Studies have found that distal EC3 synaptic input is required for plateau initiation in CA1 pyramidal neurons^{50,75}. However, proximal input can also contribute, probably by reducing the amount of EC input required to reach threshold^{35,50,75}.

This point will be raised again below, when we discuss synaptic credit assignment in CA1.

Hippocampal PF formation and a new form of plasticity

The findings discussed above raise intriguing questions about the cellular and circuit-level mechanisms of the experience-dependent adaptive change in hippocampal area CA1. To examine this, investigators used whole-cell patch-clamp techniques to record intracellular membrane potential (V_m) from CA1 pyramidal neurons as head-fixed mice learned to run laps on a treadmill for water rewards located at a particular position^{53,75–83}. Fluctuations in V_m suggest that even CA1 pyramidal neurons that do not yet have a PF receive synaptic input at kilohertz rates, although no obvious, consistent depolarizations are observable⁷⁸ (Fig. 2e). Evidence indicates that this condition is achieved by an amazing balance between the low-weight excitatory synaptic input and local network inhibition ($E-I$ balance) that leaves the V_m fluctuating around what is essentially the resting V_m recorded under in vitro conditions, when these same neurons receive very little to no synaptic input⁷⁸. Eventually, in some of these initially silent cells, the initiation of a plateau was observed to occur at a particular location while the mouse was running during a given trial. On the following trial, a fully formed PF was observed near the location where the plateau was initiated (Fig. 2e–h). The selective AP firing associated with the PF was found to be produced by a slow (on the seconds timescale) ramp of V_m depolarization that consistently drove high-frequency APs at the same position on the track (Fig. 2f,h). That this depolarization was not present before the plateau suggests that the initiation of a plateau rapidly enhanced the strength of the excitatory synaptic input, such that it was sufficient to generate the approximately 10 mV depolarization that underlies CA1 PF activity. In this case, the increased weights of the excitatory synapses break the $E-I$ balance that was present in the silent cell before the plateau, producing the ramp of V_m depolarization that underlies PF firing^{53,75,78–84}.

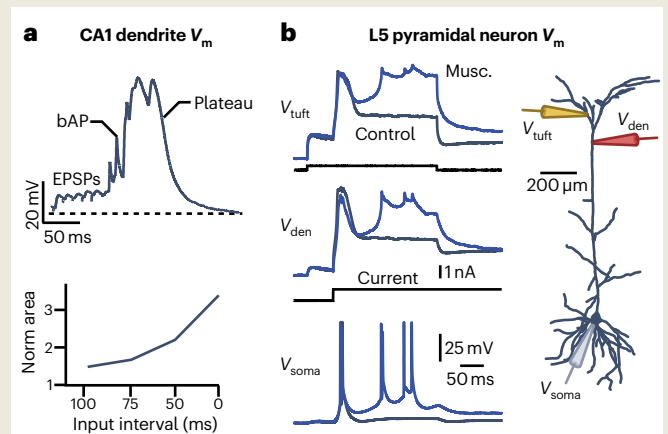
Subsequent studies both in behaving mice and in hippocampal slices revealed that a new form of synaptic plasticity driven by plateau initiation, instead of APs, produces the rapid synaptic weight changes that underlie PF formation^{53,75,78–90}. Indeed, the repetitive generation of high-frequency trains of APs (100 Hz) did not cause a significant change in the V_m when applied to CA1 neurons in behaving mice⁷⁴. The synaptic plasticity driven by plateaus has several interesting characteristics. First, this plasticity is capable of rapidly producing synaptic weight changes that generate the large V_m depolarization (~10 mV) after as few as a single plateau initiation, indicating that this is a robust form of synaptic plasticity capable of underlying the single-trial changes needed for one-shot learning (Fig. 2e–h). Next, the plasticity affects inputs covering many seconds of time around the plateau and, furthermore, it affects synapses that were active before the plateau further back in time than those that came after it, such that the plasticity is asymmetric in time, projecting further backward than forward (Fig. 2g). This seconds-long time course and asymmetry are important, because they cause a predictive shift of about 10 cm in the location of PFs with respect to the location where the plateau was initiated (Fig. 2h). These are the three signatures of this plasticity: (1) plateaus abruptly produce PFs, (2) the PFs shift back in space and (3) the plasticity that underlies this process is operating on the seconds timescale^{53,75,78,80–93}. In light of this last feature, the plasticity was named BTSP.

Additional experiments revealed that BTSP is bidirectional, meaning that both potentiation and depression components are simultaneously activated by the plateau^{81,86}. This property was first observed when experimenters used current injection to initiate plateaus outside the existing PFs of CA1 neurons that were already PCs. This plateau initiation caused the PF to shift to near the location of the newly initiated plateaus, and subtraction of V_m before from the V_m after the plateau revealed both a hyperpolarization (depression) at the initial location of the PF and a depolarization (potentiation) around the location of

BOX 1

Dendritic calcium plateau potentials

- Electrogenesis:** Voltage-gated ion channels in neuronal dendrites²⁸ allow dendrites to function as more than passive filters of synaptic input. In fact, the distal apical dendrites of pyramidal neurons possess a powerful spike generator that is independent of that found in the axon. The large-amplitude (~60 mV in dendrites), potentially long-duration (>100 ms) regenerative voltage spikes initiated in the apical dendrites are known as dendritic calcium plateau potentials^{31,34–36,40–45,155–160}. In pyramidal neurons, plateaus are essentially APs whose long duration reflects the slow inactivation kinetics of the voltage-gated Ca^{2+} channels and NMDARs that primarily support the electrogenesis as well as the relatively high activation voltage (Kv channels) and slower activation kinetics (SK and BK channels) of dendritic K^{+} channels that underlie repolarization^{156,157}. Unlike axonal APs, plateaus are not binary signals but are instead analog signals with a duration that depends on other factors such as dendritic V_m depolarization and modulation state (see below)^{42–45}. These large global voltage signals should not be confused with more local dendritic electrogenesis mediated by dendritic Na^{+} spikes or NMDA spikes, whose effects are generally restricted to individual dendritic branches by either their transient nature (Na spikes) or their dependence on active synaptic inputs (NMDA spikes)^{38,39,158,159}. However, these 'local' events, particularly NMDA spikes, can contribute inward current that enhances the duration and propagation of plateaus^{42,43}. Although the focus here is on hippocampal CA1 and neocortical layer 5 (L5) pyramidal neurons, other neurons have been reported to support Ca^{2+} -channel-mediated dendritic electrogenesis (for example, Purkinje cells, thalamocortical neurons, motor neurons, striatal and amygdala neurons)^{41,161–164}.
- Propagation:** Once initiated, the frequency dependence of passive dendritic filtering (low frequencies are filtered less than high frequencies^{165,166}) allows plateaus to propagate throughout the neuron much more effectively than more transient Na^{+} APs. They have, in fact, been observed to fully propagate throughout the entirety of CA1 pyramidal neurons¹⁶⁷. On the other hand, propagation of plateaus has been reported to be impeded by active membrane conductances such as hyperpolarization-activated channels (HCN or Ih) and certain Kv channels^{40,43,44}. In addition, computer modeling suggests that the presence of spatially distributed synaptic input to the apical dendrites might also enhance propagation¹⁵⁶.
- Inputs:** In both types of pyramidal neuron, excitatory input to the distal tuft regions appears to be required for the generation of long-duration plateaus^{40,42–44,50,66,75}, although this is more complex in the very long apical dendrites of L5 cells. These neurons possess an initiation zone about halfway along their apical trunk that generates brief calcium spikes, whose duration is then extended by excitatory input to the tuft^{34,43,44,158}. This effect is likely to result from inactivation of certain Kv and HCN channels that control plateau repolarization^{43,44} and allows more proximal synaptic input, which may be involved in trunk spike initiation, to interact with distal tuft input to alter plateau duration. The more compact dendrites of CA1 neurons do not appear to have this two-stage process, although more proximal synaptic input has also been observed to reduce the amount of distal input needed to evoke plateaus^{50,75}.
- Regulation:** The initiation and duration of plateaus are negatively regulated by dendritic inhibitory input^{102,168–170}. The disinhibitory action of interneuron-targeting interneurons (expressing vasoactive intestinal peptide) plays an important role in properly regulating these interactions^{69,104}. Finally, many of the ion channels that mediate dendritic electrogenesis are altered by the major neuromodulators. Chief among these are norepinephrine and acetylcholine, which downmodulate Ih and upregulate Ca^{2+} channels, respectively, to enhance initiation, duration and propagation of plateaus in both brain regions^{45,156,171–176} (Box Fig. 1). This arrangement allows brain state to control the integration mode of the apical dendrites, perhaps enabling arousal, attention and other factors to regulate the functions associated with plateaus.
- Functions:** The enhanced propagation of plateaus allows them to drive high-frequency bursts of AP output from the axosomatic regions of pyramidal neurons^{31,34,50,75,77,98,177}. Because of this, plateaus mediate gain modulation operations in which output is multiplicatively scaled by input initiating plateaus¹⁷⁴. This allows the formation of the complex mixed-feature selectivity and enhanced sensory perception that are required in task learning and potentially other cognitive functions^{42,178–185}. In addition, the effective propagation of long-duration plateaus throughout the entire dendritic arbor allows them to rapidly and effectively induce plasticity at any synapse within the neuron. This powerful global signal may underlie one-shot learning^{53,75}.



Box Fig. 1 | Dendritic plateau potential regulation by synaptic input and cholinergic modulation. **a**, Upper, dendritic V_m with plateaus in CA1 pyramidal neurons in vitro. Synaptic input (excitatory postsynaptic potentials or EPSPs), backpropagating action potentials (bAPs) and plateau potential (plateau). Lower, plateau area is enhanced by coincident proximal and distal excitatory input (input interval). Norm., normalized. **b**, Left, simultaneous V_m recordings from two dendritic sites (V_{tuft} , V_{den}) and soma (V_{soma}) of L5 pyramidal neurons in vitro show the impact of a muscarinic acetylcholine receptor agonist (musc., blue) on plateau duration and somatic output (control, gray). Current to dendrite sites is indicated in black. Right, L5 pyramidal neuron with recordings sites. Reproduced with permission from ref. 45, Elsevier.

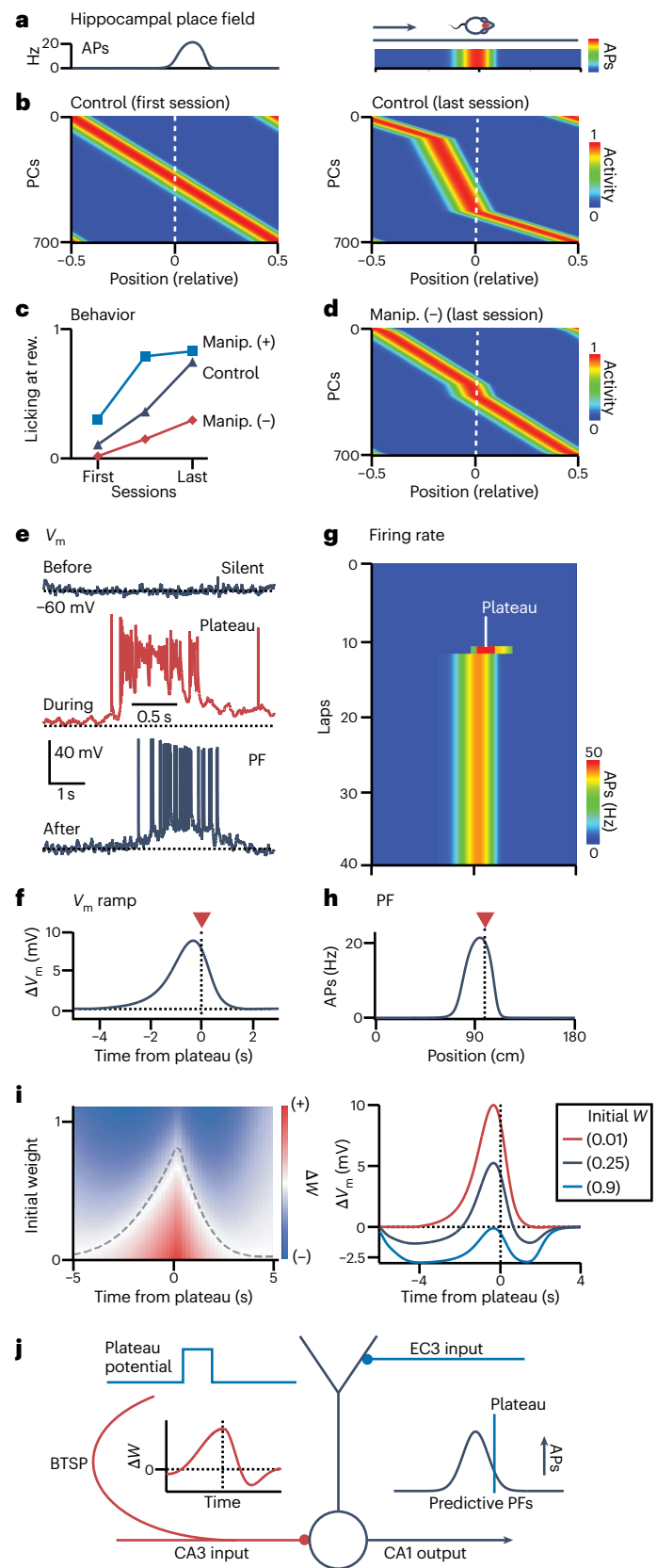
the induced plateau (Fig. 2i). This combination of potentiation and depression creates a nicely balanced plasticity that does not require any additional limiting mechanisms to keep it from either exploding or collapsing⁸¹. Instead, the BTSP rule generates a desired synaptic weight for every active synapse in the neuron that depends on the time interval between the input and the plateau. The actual weight change produced is then a function of this interval and the current weight of the synapse at the time of the plateau (Fig. 2i). The end result is that, regardless of the current weights in the neuron at the time of the plateau, BTSP will produce a PF-shaped V_m ramp that generates a new PC. In sum, plateau potentials drive BTSP to adjust the weights of CA3 inputs⁸⁵ to, in a single trial, form new PFs (Fig. 2j). In the BTSP learning rule, the postsynaptic APs of Hebbian plasticity have been replaced with a single plateau and, as discussed below, this may provide a mechanism by which experience can rapidly shape hippocampal activity to improve behavior.

Given the importance of area CA3 in hippocampal function, the question arises as to whether BTSP is involved in PF formation in this region, too. In contrast to area CA1, area CA3 is a highly recurrent network in which attractor-like dynamics appear to generate reliable and consistent PC sequences. PF activity in this network has recently been observed to be produced by BTSP at CA3–CA3 recurrent synapses⁹⁴. Interestingly, although BTSP here is similarly robust, bidirectional and covers seconds of time, the actual time course is symmetric in that it extends approximately 5 s in each direction around the plateau. This property produces the appropriate positive feedback needed for stable persistent activity in similarly tuned PCs, generating stable attractor dynamics and enhancing memory storage capacity in area CA3 (ref. 94). The symmetric plasticity rule in CA3 also suggests that modifications to the biochemical elements involved in producing BTSP can be adjusted to the needs of the network.

In summary, it has been proposed that BTSP is the main mechanism for producing a variety of PFs in both CA1 and CA3 hippocampal areas^{53,66,75,78,80–83,85–87,89,92–95}. In both cases, plateaus are a robust plasticity-induction mechanism that rapidly adjusts all the weights of synapses that were active within an approximately 10-s time window around the plateau. The exact properties of the plasticity rule can be modified to best support network function, such as to produce either predictions in CA1 or stable network dynamics in CA3.

Fig. 2 | PCs, hippocampal adaptive change and BTSP. **a**, Left, PC activity with location-selective AP firing on a linear track. Upper right, mouse on track; lower right, heat map of single PC activity (warm colors indicate higher firing rates). **b**, Heat maps of PC population (700 PCs) on initial environment exposure, sorted by PF location (left). Dashed line at reward site. Note uniform PC distribution. Right, same PC population after several sessions shows increased number of PCs with PFs near reward site (overrepresentation). **c**, Behavior during control sessions (black) and two manipulations affecting overrepresentation (blue, decreased inhibition; red, increased inhibition). **d**, Increased dendritic inhibition blocks overrepresentation formation. Data indicate a causal relationship between PC population activity and behavioral performance, with overrepresentation as an adaptive change⁶⁹. PFs and heat maps are schematized. **e**, Properties of BTSP and PF formation in single neurons. V_m traces before (silent, top), during (plateau, middle) and after (PF, bottom) plateaus that induced new PFs. Dashed line is at -60 mV. **f**, Change in V_m caused by plateau induction of BTSP (V_m ramp). Note change spans ~ 10 s and projects further backward than forward. **g**, Schematized heat map of position-dependent firing rate (AP, Hz) for multiple laps showing abrupt formation of PFs following plateau initiation. **h**, Average firing rate for trials after plateau initiation. Red arrowheads and dashed lines indicate plateau location in **f–h**. **i**, Left, BTSP induced synaptic weight change (ΔW) profile. ΔW is a function of the synaptic input and plateau initiation time interval and initial synaptic weight. Dashed line indicates regions of no change. Right, V_m changes for slices through a three-dimensional plot at indicated initial synaptic weights that span all potentiation (red) to all depression (blue) with a mixture at mid-level (black). **j**, Schematic of pyramidal neuron receiving EC3 input to distal dendrite compartment, where a plateau is initiated to drive BTSP to reshape synaptic weights of CA3 inputs to produce predictive PFs as CA1 output.

Likewise, the exact function of the EC instructive signals could vary in the different regions, where they could generate either unique, experience-dependent PC sequences in CA3 (ref. 94) or PC representations with variable spatial densities in CA1 (ref. 88; see next section). Currently, the properties of EC input to the distal integrative compartments of CA1 and CA3 are less well understood, and future experiments



BOX 2

Evidence of BTSP involvement in hippocampal adaptive change formation

1. Physiological evidence that BTSP induction can rapidly form new PFs

For: Naturally occurring and experimentally induced plateaus rapidly form new PFs by bidirectionally changing the V_m depolarization profile over the seconds timescale associated with BTSP^{53,75,78–86,94}

Caveat: Only indicates BTSP is sufficient for PF formation

2. Frequent observations of BTSP signatures during learning-related representation formation

For: Observations that most PCs appear abruptly, shift back in space relative to initiating event (large signal thought to be plateau) and produce PFs whose width is correlated with running speed near initiation^{87–93,95,96}

Caveat: PFs that do not show obvious BTSP signatures are also less frequently observed

3. Two pharmacological manipulations of BTSP inhibit learning-related representation formation

For: Application of NMDAR or R-type Ca^{2+} channel antagonists to the recording field reduced new PF formation and blocked the development of an overrepresentation of salient regions^{50,53,66}

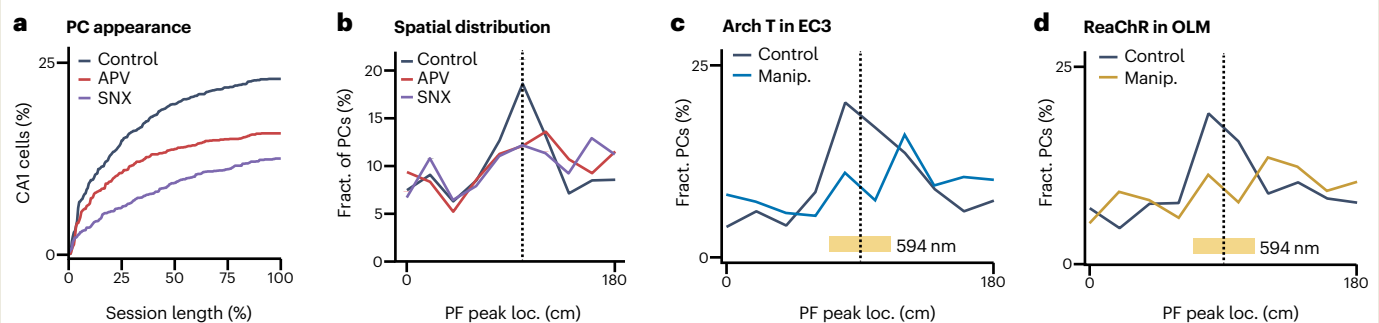
Caveat: Non-specificity; other plasticity types are NMDAR-dependent and may depend on R-type channels

4. Multiple manipulations of microcircuit elements controlling BTSP induction inhibit learning-related representation formation

For: Reducing plateau initiation in CA1 by either lowering EC3 excitatory drive to distal dendrites or enhancing dendrite targeting inhibition reduces new PF formation and interferes with the development of learning-related population activity^{66,69,72,75,96,103–105}

Caveat: Unknown off-target circuit effects

Summary: Multiple lines of experimental evidence support BTSP being the major mechanism mediating learning and memory in the hippocampus. Questions remain as to whether BTSP is the sole synaptic plasticity mechanism involved in hippocampal PF formation.



Box Fig. 2 | BTSP in adaptive change formation. Manipulations of BTSP alter overrepresentation of reward location. **a**, Progressive increase in PC number in representation across behavioral session for control (black) is reduced by two pharmacological inhibitors of BTSP (APV, red; SNX482, blue). **b**, These agents also alter PC representation. **c**, Inhibition of EC3 input (blue). **d**, Activation of OLM interneurons (orange) alters PC spatial distributions when compared to control (black)^{66,105}.

should explore, among other things, the functional role of plasticity at entorhinal to hippocampal synapses.

BTSP and adaptive changes

Is BTSP involved in producing the remarkable adaptive changes in area CA1 that are required for hippocampal-dependent learning? Several groups have reported that when CA1 PCs initially appear during the formation of new environment-specific adaptive population activity, most show the three signatures of BTSP^{87–95} (Box 2). In addition, local application to the recording field of two different known pharmacological inhibitors of BTSP significantly reduced the number of PFs that appeared over the session and eliminated the overrepresentation of the salient regions⁶⁶ (Box Fig. 2). The above evidence was interpreted to indicate that BTSP is responsible for the adaptive changes in CA1 population activity that are required for an animal to learn the appropriate behavior just by interacting with the world (Box 2).

Having said this, many investigators find that a minority of CA1 PCs (usually less than 25%) do not necessarily show the above hallmarks of BTSP^{87,89,91–93,95}. Some of these PCs have PFs on the very first exposure to the new environment, while others appear later but without an

obvious backward shift or indication of plateau initiation. This could indicate that although BTSP may be the major plasticity form shaping learning-related population activity in the hippocampus, it might not be the only one. However, it is also possible that PFs, particularly weaker fields or even subthreshold depolarizations, can appear without any additional plasticity if enough active inputs already have appropriate weights set during past learning. Indeed, it seems unlikely that the hippocampal weight matrix associated with active synaptic inputs is ever somehow a complete tabula rasa, even when an animal is experiencing a new environment. Thus, if BTSP is the main plasticity mechanism in the hippocampus, then the most straightforward answer would be that BTSP is also mostly responsible for these initial PFs too. We further discuss the theoretical role of synaptic weights set by past learning below (see 'Credit assignment in BTSP').

Neuronal circuit elements involved in controlling BTSP

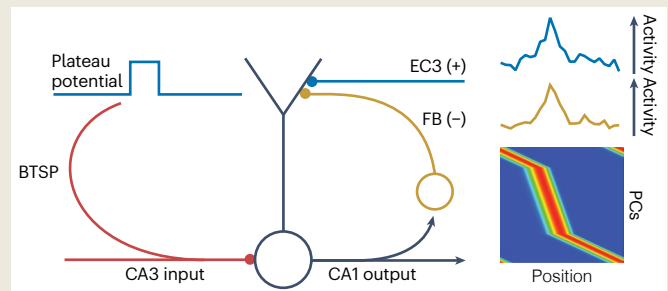
Additional experiments examining the roles of particular neuronal circuit elements further confirmed the role of BTSP in learning-related population activity changes in CA1. First, it was found that delimited optogenetic inhibition of the excitatory input from the EC3 during

BOX 3

Circuit elements controlling BTSP and adaptive change formation

Neuronal circuit elements appear to produce predictions of an environmental variable, perhaps value, in the population activity of hippocampal area CA1. In this scheme, the EC3 delivers an excitatory input functioning as a population-level target signal related to the current spatial distribution of environmental value, to the distal dendrites of all CA1 pyramidal neurons^{66,75,96} (Box Fig. 3; blue), where it is compared with a representation of the actual CA1 population activity, probably provided by the OLM feedback (FB) inhibitory interneuron population activity^{69,72,103–105} (Box Fig. 3; orange). The probability of a cellular-level error signal will be increased in those pyramidal neurons where EC3 excitation exceeds feedback inhibition, and this local error signal is mediated by plateau potentials. Plateaus are robust inducers of BTSP, and this synaptic plasticity adjusts the weights of the CA3 excitatory inputs (Box Fig. 3; red), changing CA1 population activity until it matches the EC3 target. At this point, both the number of PCs in the representation and the activity of feedback inhibitory elements will be elevated at the most salient regions of the environment (Box Fig. 3; CA1 population heat map). The local feedback inhibition counters the excitation from the EC3 target signal (Box Fig. 3; orange and blue traces), reducing plateau initiation and any further synaptic plasticity, and thereby effectively halting any additional changes in the population activity as the appropriate adaptive change has been achieved. Altogether, it appears that various neuronal circuit elements control a form of synaptic

plasticity named BTSP to produce, within hippocampal area CA1, the adaptive changes that improve behavior. This may represent the main circuit-level and cellular-level mechanisms of learning in the hippocampus.



Box Fig. 3 | Network scheme for learning in CA1. Excitatory input from EC3 (blue, upper right) provides target activity pattern (blue activity; upper right) that is compared in the distal apical dendrites with the actual CA1 population activity provided by OLM interneurons (orange; FB–; middle right, orange activity late in session). Plateau initiation (blue, upper left) plateau induces BTSP at CA3 excitatory inputs (red), altering CA1 network activity to form the adaptive change (schematized PC heat map with overrepresentation from late in session; bottom right).

learning significantly reduces the overrepresentation of salient regions, presumably by lowering plateau initiation and thus BTSP induction in CA1 (refs. 66,96; Boxes 2 and 3). This fits with the idea that EC3 input is a type of teaching signal that is, in this case, directing BTSP to increase the number of PFs at important regions of the environment.

The next question to be addressed was regarding the form of the EC3 teaching signal (for example, an error signal). The first step was to examine the activity of the EC3 axons in CA1, where it was found that there was a strong peak in average velocity-corrected EC3 neuronal activity at the salient regions of the environment, as would be expected for a teaching signal^{66,97,98} (Box 3 figure). Interestingly, this peak in EC3 activity around important regions remained constant for the entire session, instead of, as would be expected for a standard error signal, decreasing as CA1 activity adapts to the environment (that is, as the overrepresentation builds). In support of this idea, the similarities between the CA1 and EC3 average population activities started low at the beginning of the session but ended high as the CA1 overrepresentation built during learning of the new environment. Thus, it appears that, during the learning session, CA1 population activity is changing to match that in EC3, suggesting that the EC3 input is not an error signal but instead a target or desired activity pattern that tells the CA1 to put three times more PCs at salient regions. The idea that EC3 provides CA1 with a target signal is interesting, because target signals are, in theory, quite powerful at directing learning in complex neuronal networks^{15,99}. Thus, the use of experience-dependent targets to generate neuron-specific error signals that engage the BTSP learning rule could solve the credit assignment problem in hippocampal area CA1 (ref. 81). In the end, this powerful form of directed learning is made possible by the second integrative compartment produced by both the passive and active properties of pyramidal neuron dendrites.

Finally, there exists an inhibitory feedback element within the CA1 microcircuit that may have an important function in this process (Boxes 2 and 3). A set of somatostatin-positive interneurons named oriens/lacunosum moleculare (OLM) neurons receives excitatory input

almost exclusively from CA1 pyramidal neurons, and they target their axons nearly exclusively to the distal apical regions of CA1, where they make inhibitory synapses with the dendritic tufts of CA1 pyramidal neurons^{100–102}. They are thus a dendritic inhibitory element that is driven by feedback excitation from CA1 pyramidal neurons and, as such, could provide inhibitory control over the initiation of dendritic plateaus that mirrors PC population activity. Consistent with this idea, optogenetic inhibition of these somatostatin-positive interneurons in behaving mice increased burst firing in CA1 pyramidal neurons¹⁰². In addition, recordings from OLM interneurons confirmed that their activity followed that of the PC population, as OLM activity at the salient regions of the environment increased during learning, just as is seen for the PC population¹⁰¹. Further, experiments using selective optogenetic manipulations of feedback inhibitory interneuron population activity during learning altered the standard development of overrepresentations of salient locations in the PC population activity^{69,72,103–105}. Thus, the activity of feedback interneurons could function as a brake on the formation of new PFs during the development of learning-related CA1 PC representations. Altogether, it appears that various neuronal circuit elements control a new form of synaptic plasticity called BTSP to produce within hippocampal area CA1 the adaptive changes that improve behavior. This may represent the main circuit and cellular-level mechanisms of learning in the hippocampus.

A role for BTSP in memory formation and retrieval

We can now examine the second stage mentioned earlier: the stabilization of synaptic weight changes that reflect learning (that is, memory formation). The process of memory formation has its own complexity known as the plasticity–stability tradeoff or dilemma^{106–109}. This tradeoff can be explained as follows: to profit from past learning, beneficial information accumulated by animals during a behavioral episode should be retained and later retrieved during similar experiences^{59,60,106,107}. Thus, some amount of the synaptic weight changes should be stabilized such that they exist for the duration of a memory. However, if all or some large

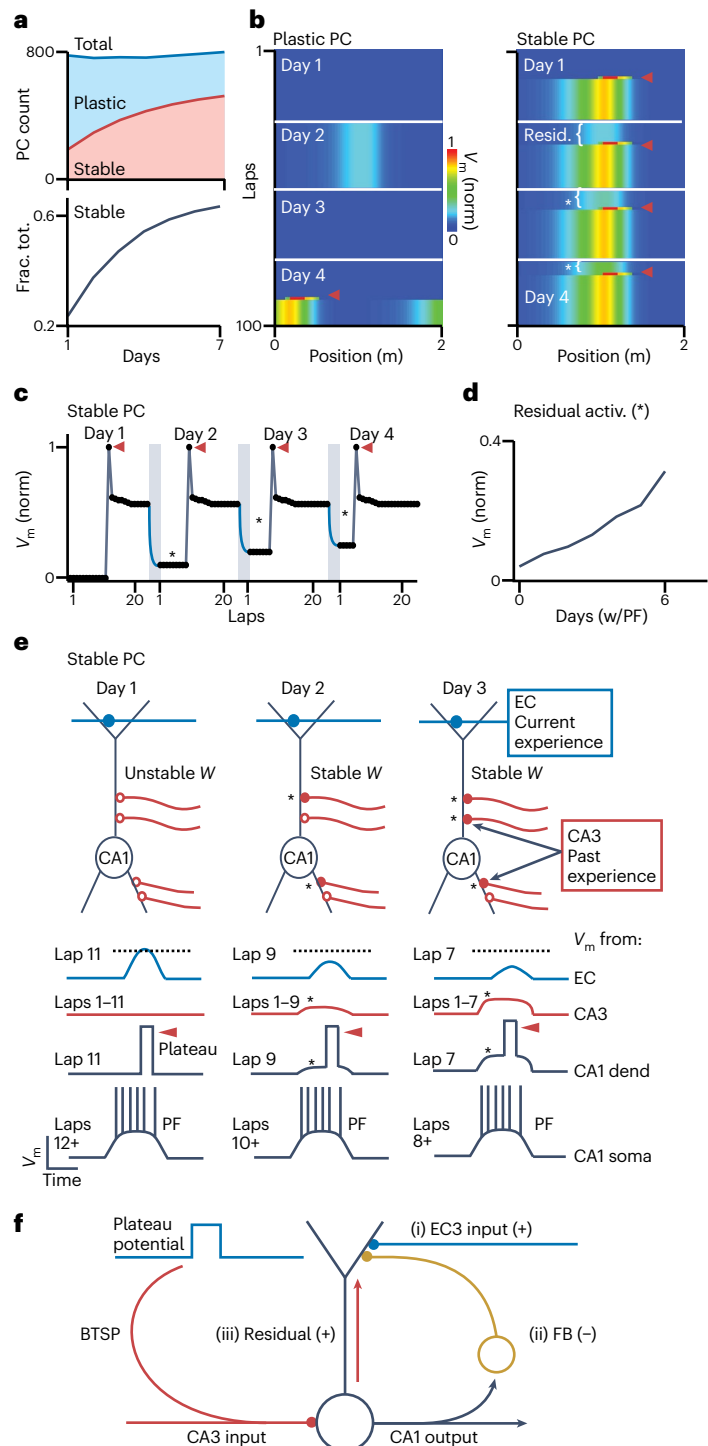
fraction of the weight modifications underlying the adaptive changes are completely stabilized, then the capability for additional learning will be reduced. On the other hand, if these synaptic changes remain fully plastic, they will be rapidly overwritten by subsequent learning, and little will be retained for the long term. Thus, a balance is needed where perhaps some component of the adaptive changes is stabilized, protected from overwriting, while the rest remain plastic for reuse in new learning.

To examine how learning-related changes in CA1 activity are maintained over extended periods (>1 week), researchers generally use optical imaging techniques to longitudinally record the activity of the same CA1 neuronal population in mice engaged in a behavioral task. Numerous labs have tracked CA1 hippocampal PC activity over days, and most of them report a progressive decrease in the similarity of population activity from a given day onward^{71,95,110–116}. In general, although most PCs join the representation only transiently, there is usually a second population of more stable PCs that maintain consistent PF activity for many days^{71,95,110–116}. Recently, one group reported that the size of the stable PC population progressively expanded nearly threefold across a week of task performance, such that by the end of a week most of the PC representation was composed of stable PCs, while there remained a lesser but still substantial component of plastic PCs newly appearing each day⁹⁵ (Fig. 3a). They also found that the stable PCs disproportionately represented task-related learned information and were retrieved earlier within a behavioral session, and the size of this population showed a strong correlation with behavioral performance.

The fact that only a subset of PCs showed strong stabilization across a week of training could be an indication of how the hippocampus handles the stability–plasticity dilemma, with only a fraction of PCs becoming stabilized in the long term (>7 days). However, this

study also reported that even the stable group of cells showed obvious signatures of new BTSP each day when robust PC activity was formed at the same location as previous days⁹⁵. That is, each day, new BTSP evoked by additional plateaus was required for a given PC to be active at the same PF location, presumably because overnight the strength of the excitatory synaptic input weights had either decayed or been overwritten (Fig. 3b–e). These data suggest that it is actually an increase in the likelihood of new neuron-specific synaptic plasticity, as opposed to extensive long-term synaptic weight stabilization, that is responsible for producing the memory. This contrasts with classical hypotheses, which hold that the prolonged maintenance of learned synaptic weights produces the long-term storage needed for memory formation and retrieval^{3,4}. However, the above evidence indicates that

Fig. 3 | PC stabilization and credit assignment in BTSP. a, Top, the number of stable CA1 PCs in the population increases across one week of experience, while plastic PCs decrease and total PC count stays constant. Bottom, the fraction of the total population of PCs that are stable increases. **b**, Schematized PC activity heat maps of V_m for a plastic PC (left; weak depolarization on day 2, stronger activity at different location on day 4 following plateau, red arrowhead) and a stable PC (right; strong depolarization at same location for all days). Note residual weak depolarization (asterisk) in laps before plateau initiation (red arrowheads) in a stable PC. Experimental days indicated (25 laps per day). **c**, V_m amplitude across days for a stable PC shows that each day a new plateau (red arrowheads) is required for strong PFs and each day the amplitude of the residual (asterisk) increases. Gray bars are ~23-h periods between recording sessions (~1 h) when overwriting and decay occur (light blue lines). **d**, The amplitude of the residual component for the population increases as PCs appear for more days suggesting a stabilization process. **e**, Sketch of PF formation for a stable cell (**b** and **c**) on day 1 (stable PC; left traces) showing depolarization from EC3 (blue) and CA3 (red) inputs with no residual input from CA3 (unstable weak synapses; open circles). Plateaus on trial 11 that induces new strong PFs requires strong EC input. PF formation on subsequent days of appearance (day 2 and day 3) showing plateau initiation on earlier trials as increasing residual depolarization (asterisks; stable synapses; filled circles) progressively lowers the level of matching EC3 input required for plateau initiation. Black lines are composite V_m from CA1 cell. Particular PF locations are hypothesized to be chosen based on a comparison between the current experience delivered by EC3 input and past experience stored in stabilized CA3 synapses. The number of stable synapses and level of stabilization is hypothesized to increase with repeated plateau initiation. The dashed line is the plateau threshold without residual depolarization to show plateaus are initiated for lower levels of EC input when residuals are present. **f**, Schematic of three main elements in dendritic computation controlling plateau initiation (upper left, blue) and BTSP induction (dark red). (i) EC3 input to distal dendrite compartment functioning as a population-level target signal (upper right, blue). (ii) Actual CA1 population activity delivered by feedback inhibition. (iii) Excitatory component mediated by stable weights set by past learning that could provide cell and location specificity for PC representation formation (red arrow; residual).



BOX 4

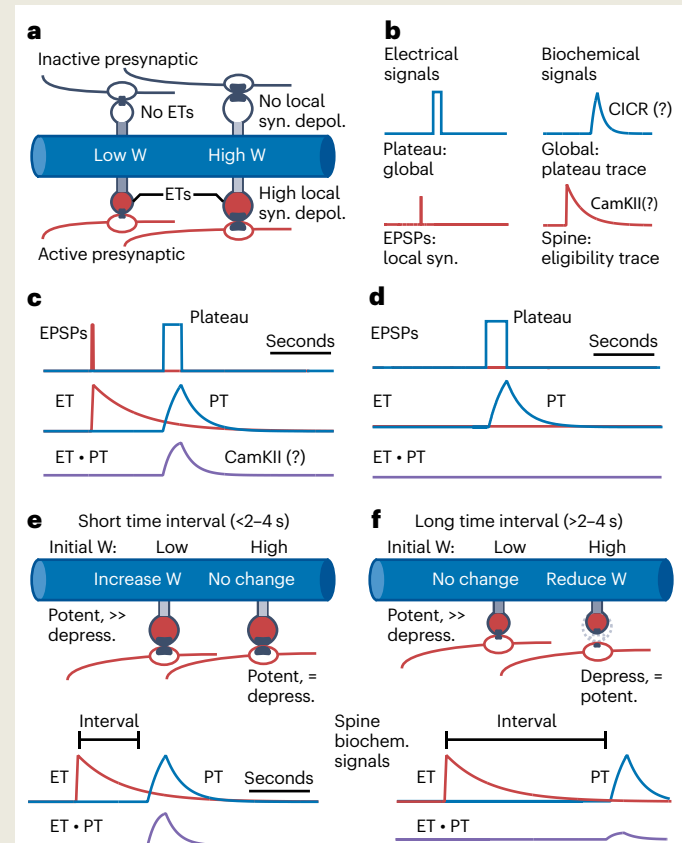
BTSP-associated signals

One of the main mysteries accompanying BTSP is the seconds-long timescale over which synaptic potentiation and depression occur. How is it possible for synaptic input and plateaus to interact when they are separated by many seconds? For these electrical signals to appropriately overlap, both may use biochemical filters that extend their time courses to cover the approximately 10 s time window over which BTSP changes synaptic weights. A filter of synaptic input is known as an eligibility trace (ET) and, although first proposed decades ago, remains incompletely described^{6,12}. In the hippocampus, the biochemistry underlying the ET should produce a trace of the synaptic input lasting 5–6 s (Box Fig. 4). If the ET is the only trace present, then only those inputs that preceded the plateau will have their weights changed. To produce the weight changes occurring after the plateau, another seconds-long trace of the plateau itself is required (plateau trace or PT; Box Fig. 4). The exact shape of the weight-change time course will be a function of the ET and PT durations and has been observed to vary from being biased in a backward direction (ET longer than PT in CA1) or symmetric around the plateau (ET and PT equal in CA3)^{6,53,86,94}. There are three other terms in the equation describing BTSP^{52,80}, and it is possible that a

different molecule could mediate each of the five different elements (ET, PT, overlap signal and two nonlinearities). On the other hand, perhaps a single reaction underlies the entire process.

Future biochemical studies should recreate as closely as possible the fundamental conditions associated with BTSP. First, these experiments should adhere to basic physiological constraints (studies performed at ~37°C; ideally, the tissue should be from adult animals, and the synapses under study should be found in normal brains, that is, tissue culture should be avoided owing to autapses, for example). Second, long-duration voltage changes mimicking plateaus should be the induction mechanism. Likewise, to identify molecules that underlie noncontingent ETs, synaptic inputs should not be closely associated with APs. Third, the time course of the induced weight changes should match that observed in BTSP, that is, it should operate on the seconds timescale as opposed to either milliseconds or even many tens of seconds. In the end, the plasticity that is under study should be one that can form new PFs in a single shot under conditions as similar as possible to those present within the behaving animal.

Box Fig. 4 | Which synapses change and how. **a**, Postsynaptic dendrite with spines and presynaptic inputs. The upper spines in the diagram receive inactive inputs (black); lower spines receive active inputs (red). **b**, Electrical signals: plateau potentials (blue, left) produce global dendritic depolarization and EPSPs (red trace, bottom left) produce local synaptic depolarization. Associated biochemical signals: PT (top right) and ETs (bottom right). CICR and CaMKII have been hypothesized to be involved. Only active inputs generate ETs (spines shaded red in **a**). **c**, Overlay of electrical (light red and blue) and biochemical signals (dark red and blue) generated by active inputs. Synapses with overlapping ET and PT (ET•PT; purple) have their weights adjusted. **d**, Same as **c** but for inactive inputs. The lack of ET yields no overlap signal and no weight adjusting. **e**, Shorter intervals between synaptic inputs and plateau produce large overlap signals. Resulting weight changes are dominated by potentiation due to subsequent nonlinear potentiation and depression processes, which are scaled by initial weights (potentiation of initially low-weight synapse indicated by increased spine size). Weight change is a function of the initial weight and the time interval (initially high-weight synapse does not change weight). **f**, Longer intervals produce smaller overlap signals and depression dominates (left). The exact magnitude of synaptic depression is a function of the interval and the initial weights (initially high-weight synapse depresses as indicated by decrease in spine size, initially low-weight synapse does not change weight). In **a**, **e** and **f**, adjustments to spine neck resistance (lighter gray shading denotes lower neck resistance) could limit weight-related changes in spine depolarization, allowing all synapses to maintain their sensitivity to input regardless of weight.



in nearly all PCs, including those that maintain stable representations across days, many synaptic weights modified during learning do not maintain their strength between sessions, suggesting that most weights in the network remain fully available for new plasticity and only a sparse set is highly stabilized. As it is theoretically advantageous during continuous learning conditions to stabilize only a fraction of learned synaptic weights, while leaving the majority available for overwriting, this mechanism may allow the hippocampus to achieve a proper stability–plasticity balance^{105–108,116–120}.

Credit assignment in BTSP

How is the likelihood of new BTSP increased in particular neurons at specific locations to produce stable PF firing across one week of behavior? The above studies also report that in the trials preceding the initiation of a plateau, many neurons exhibit a weak and inconsistent level of activity near the location of the future plateau and resulting PF^{93,95,121,122}. Whole-cell recordings in CA1 also find that in many future PCs, there can be an initial subthreshold or weakly suprathreshold level of V_m depolarization near the location of the plateau potential

BOX 5

New concepts and questions raised by BTSP

1. Learning controlled by a dendritic computation featuring:

- An instructive signal related to current experience from a higher-order brain region^{50,66,75,96}.
- Past learning embedded in synaptic weights^{93,95}.
- Regional activity levels delivered by local inhibitory elements^{69,72,88,102–104,164,165}.
- Permissive control via neuromodulatory regulation^{45,95,169,170,180}.
- Function: allows a self-supervised plasticity that may link current experience to weight adjustments yielding useful changes in neuronal population activity.
- Contrasts with concepts based on correlated AP firing between neuron pairs or an explicit neuron-specific teaching signal^{3–6}.

2. Memory via enhanced likelihood of new neuron-specific plasticity:

- Incomplete synaptic weight stabilization during or after learning^{93,95}.
- Most PCs require re-instantiation via BTSP for robust PF activity^{87,91–93,95,121,122}.
- Induction of new plasticity influenced by residual level of stable weights^{93,95}.
- Stabilization occurs on slower timescales (days) during repeated experiences⁹⁵.
- Function: solution to stability–plasticity dilemma. Past learning influences new learning.
- Contrasts with concepts based on complete reactivation of past neuronal activity via stable synaptic weights in feedforward or recurrent networks^{3,4,94}.

and resulting PF^{93,95} (Fig. 3b–e). These data suggest that synaptic input to these neurons may contain a residual level of high weights that, by producing a small amount of spatially tuned depolarization, effectively reduces the threshold for plateau initiation at particular locations, thus biasing which neurons will become PCs and at what PF location. Theoretically, these high-weight synapses could be associated with either the CA3 or the EC3 input streams, although the reduced level of spatial tuning present in the EC3 pathway suggests that input from CA3 is most likely to drive this residual depolarization. Overall, that information contained within a residue of stable synaptic weights can shape new PC generation may provide a mechanism for past learning to influence new learning (Fig. 3e).

This interesting interaction may underlie a new high-dimensional credit assignment mechanism within BTSP. To adjust each of the 6 billion synaptic weights in area CA1 toward improving behavior, all the network needs is a target signal that initiates plateaus in the correct neurons at the appropriate positions. Once this occurs, the BTSP learning rule will determine which synapses change and exactly how (Fig. 2 and Box 4). Now, because EC3 input activity to area CA1 exhibits a sizable amount of stochasticity in our rather simple behavioral conditions, and because each input axon innervates thousands of CA1 pyramidal neurons, it is not obvious how the EC3 input alone could be capable of operating as an explicit neuron-specific instructive signal, such as those thought to exist in the cerebellum^{66,123,124}. Although it is possible that under more demanding behavioral conditions⁶⁶, synaptic plasticity at the EC–CA1 synapses—in combination with enhanced firing selectivity—could allow the EC3 target to pick the correct PCs and PFs^{15,99}, it seems likely that another factor is also involved. The above

evidence suggests that this other component could be the synaptic weights already present within neurons that were set and stabilized during past learning. Thus, credit assignment in CA1 could be a function of matches or similarities in the properties of the current experience delivered by input from the EC3 population target and past experiences stored in the fraction of stable weights in the active CA3 synaptic input. This combined use of a more global or population-level target signal with a local, neuron-specific, biasing signal appears to be the basis of high-dimensional credit assignment in CA1. In addition, the use of information stored in stable weights (memory) to influence new plasticity in associated synaptic inputs could mediate computations such as generalization, which here is an abstraction that finds useful common features of multiple experiences or episodes. Future experiments should explore the new credit assignment mechanism in CA1 and determine whether BTSP supports generalization in the hippocampus via this algorithm. Finally, investigation of the theoretical capabilities of BTSP has had an encouraging start^{56,94,125–132}, and this should motivate continued exploration both in terms of its capacity to function as a general learning and memory mechanism and toward potential applications in biologically inspired artificial neural networks.

Molecules and machines involved in BTSP

Many interesting but unknown molecular signals and subcellular machines are needed to implement BTSP (Box 4). There have been, however, only a few studies exploring different molecules in BTSP; thus, the molecular mechanisms underlying it are poorly understood. One molecule thought to be involved in BTSP is the NMDA receptor (NMDAR)^{53,66}, and these Ca²⁺-permeable glutamate receptors appear to be well suited for the first step in transducing synaptic input into the biochemical activity that could function as an eligibility trace. Yet here their role is somewhat different than the classical presynaptic and postsynaptic activity coincidence detector originally proposed^{17,133}. The eligibility traces in BTSP are produced without postsynaptic APs (termed noncontingent ETs^{17,134}); thus, there is no coincidence to detect. This raises the question of how it is possible for synaptic input alone to produce enough depolarization to relieve the Mg²⁺ block of NMDAR. Past studies indicate that this can be achieved by dendritic spines with high neck resistances^{135–137}. However, to avoid voltage saturation and loss of driving force, such spines may require an adjustable neck conductance that is linked to synaptic conductance such that these values change proportionally. What controls neck conductance and how it is linked to synaptic weight are unknown.

Another molecule recently examined was α -calcium-calmodulin-dependent protein kinase II (α CaMKII), which is thought to be involved in synaptic modifications produced by traditional long-term potentiation (LTP) and long-term depression (LTD)^{138–144}. α CaMKII is activated by synaptic transmission-mediated calcium influx, and its subsequent phosphorylation is thought to outlast the initial triggering event by several seconds and thus was hypothesized to mediate the ET. To examine this, one group performed whole-cell in vivo and in vitro recordings in CA1 pyramidal neurons from mice engineered with a point mutation at the autophosphorylation site (p.Thr286Ala), which causes accelerated signaling kinetics^{141,142}. However, instead of changing BTSP kinetics, this mutation severely diminished BTSP amplitude⁸³. Another recent study reported that several different manipulations of α CaMKII activity also blocked BTSP in vitro¹⁴³ as opposed to changing its kinetics, although there were many aspects of this plasticity that suggested it was not actually BTSP that was under study here. Although the above studies suggest a role for α CaMKII in BTSP other than functioning as an ET, the uncertainty surrounding the mechanism by which α CaMKII affects synapse strength allows the possibility to remain.

An additional mechanism thought to be involved in BTSP is calcium-induced calcium release (CICR) from internal storage sites such as the endoplasmic reticulum. In neurons, the endoplasmic

reticulum is a complex organelle that extends throughout the apical dendrites and may be involved in signaling to the nucleus^{144–147}. BTSP has been found to require L-type Ca^{2+} channel activity, and the dendritic endoplasmic reticulum appears to interact with clusters of L-type Ca^{2+} channels on the plasma membrane^{53,147}. A recent study using manipulations of Ca^{2+} sequestration found evidence that CICR may underlie the PT, as the duration of BTSP was enhanced when sequestration was inhibited⁸⁹. Other preliminary in vitro studies in neocortical regions have found evidence that CICR could act as both the ET and PT simultaneously¹⁴⁸. The above studies suggest that Ca^{2+} release and associated organelles could be heavily involved in BTSP.

In the end, these are simply examples of potential molecules and subcellular machines that may be involved in synaptic weight adjustment by BTSP. Of course, there are many other molecules and organelles that have been identified in past studies of traditional LTP and LTD^{140–146}, which raises the question: is BTSP different from traditional LTP and LTD? To start, as BTSP is induced by long-duration dendritic plateau as opposed to coincident AP activity, the induction properties of BTSP are obviously different from those of Hebbian forms of plasticity. Further, synaptic potentiation and depotentiation (depression) are expressed simultaneously with plateau initiation in BTSP, and this contrasts with the ideas expressed in the Bienenstock–Cooper–Munro model of synaptic plasticity where the direction of the weight change is a function of AP firing rates and perhaps concomitant changes in intracellular calcium concentration^{149,150}. Despite these differences in physiological signals or theoretical rules, there may be a good deal of overlap in the molecular and cellular biological basis of the two forms of synaptic plasticity. As many of the past biochemical studies of synaptic plasticity used induction protocols that featured large and sustained levels of postsynaptic depolarization, they may have produced similar biochemical cascades to BTSP^{16,17}. Thus, the best strategy for exploring the molecules involved in BTSP may be to start testing those for which there is sound evidence for a role in standard LTP and LTD.

Summary and final thoughts

Recent studies suggest that BTSP functions to rapidly generate new hippocampal PFs and appears to be heavily involved in producing the experience-dependent hippocampal population activity patterns required for learning. The fundamental properties of BTSP mediate the above functions (Box 5). The relevant properties are a strong, bidirectional form of plasticity that affects synaptic weights over many seconds of time, driven by dendritic plateau potentials initiated, at least in part, by an instructive input from a higher-order brain region (EC). In addition, BTSP's apparent involvement in the daily regeneration of stable PF activity suggests that it provides a nonstandard mechanism for memory storage and retrieval in the hippocampus that mitigates the need for widespread synapse stability. The potentially new credit assignment procedure at work in BTSP may also allow hippocampal networks to form both new memories of specific behavioral episodes and abstractions, such as generalization that finds similarities among different experiences¹⁵¹. Given the above, BTSP and associated circuit elements could represent a general learning and memory mechanism that is present in many brain regions in addition to the hippocampus.

Thus, it is important for future experiments to explore whether other areas, particularly neocortical regions that contain similar circuit motifs, possess synaptic plasticity forms that are driven by plateau potentials rather than AP output^{152–154}. One would expect that the specific properties of such new forms of plasticity would be adjusted to the particular needs of a given network, just as different subregions of the hippocampus use different forms of BTSP. Finally, in light of BTSP's involvement in experience-dependent shaping of hippocampal population activity, future experiments exploring the molecular and cell biological elements of BTSP could shed light on the most fundamental mechanisms of learning and memory formation. Indeed,

dendritic plateau potential-driven synaptic plasticity, as first described in the hippocampus, could provide a model for future explorations of how brains learn and remember that could range from the systems and cognitive levels down to the basic biochemical building blocks of learning and memory in mammalian brains.

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Competing interests

The author declares no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Jeffrey C. Magee.

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